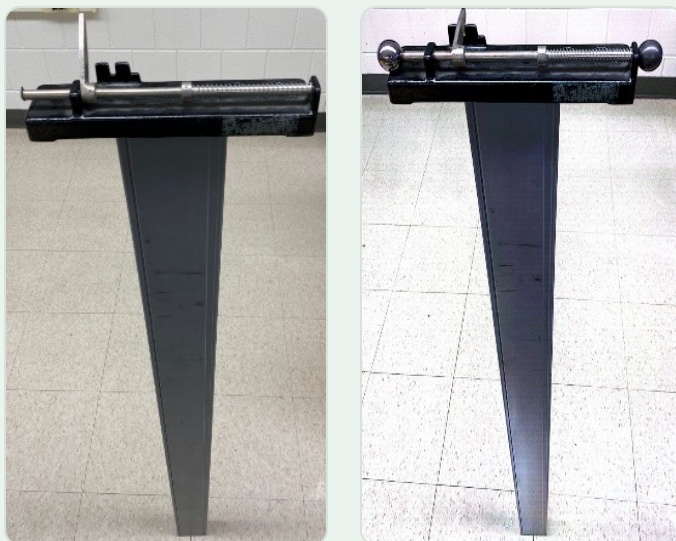


Freefall vs. Horizontal Launch

SYNOPSIS

Picture a cannonball fired from the top of a tower — the cannon aimed perfectly horizontally. At the exact instant it fires, a second cannonball is simply dropped from the same spot. The fired ball travels some horizontal distance before it lands; the dropped ball falls straight down. If you timed both balls, would one hit the ground before the other — and if so, which one — or would they land at the very same instant? This note works through a Claim–Evidence–Reasoning (CER) argument to find out, then tests it against a real demonstration.



The launcher-drop device: one steel ball is pushed horizontally while a second is released straight down, at the exact same instant.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1** Predict, and then justify in Claim–Evidence–Reasoning form, whether a dropped ball and a horizontally launched ball released together from the same height reach the ground at the same instant.

PE2 Sketch the velocity vectors at equal time intervals on trajectories A (dropped), B (launched horizontally) and C (the same launch with no gravity), getting the direction and the relative length right at every mark.

PE3 Argue from the spacing of the marks that v_x is constant (equal spacing in equal times) while v_y changes at a steady rate (spacing that grows by a fixed amount).

PE4 Write the velocity of a horizontally launched projectile as a vector, $\vec{V}(t) = v_{0x} \hat{i} + (-9.8t) \hat{j}$, and state what each term does — and does not — depend on.

PE5 Calculate the fall time of a horizontally launched projectile from the drop height alone, and explain why the launch speed does not enter the calculation.



Evidence: what you already have

Before you commit to a claim, take stock of what you've already seen — even if it turns out to be misleading. That's part of building good evidence:



Two coins off a table

Flick one coin off the edge of a table while a friend drops a second coin from the same height, at the same instant. Which do you expect to hit the floor first?



Movie physics

In action movies, a car driven off a cliff often seems to hang in the air before it drops. Is that really how gravity works?



A thrown ball

You've watched a ball thrown sideways arc downward through the air. Does it really spend more time falling than one just dropped from the same height?

In this note, we work through

I. What's your **claim**: do the two balls hit the ground at the same time, or does one beat the other?

- II. What **evidence** or experience do you already have that's relevant — and can that intuition be wrong?
- III. What do you actually **observe** when both balls are released from the launcher–drop device at the same instant?
- IV. How do the velocity vectors on trajectories A, B, and C **explain** what's happening in each direction?
- V. Why does the hypothetical no-gravity path C look the way it does?

Claim — predict first: a ball is dropped from a tower at the same instant a second ball is fired horizontally from the same height. Ignoring air resistance, which one hits the ground first?

The dropped ball lands first.

The launched ball lands first.

They land at exactly the same time.

It depends on how fast the ball is launched.

Observe: the launcher–drop device

The device pictured above holds two steel balls at the same height. A spring-loaded arm pushes one ball out horizontally at the exact instant it releases the second ball to drop straight down — no reaction time, no guessing about "at the same time." Watch (and listen) as your instructor fires it.

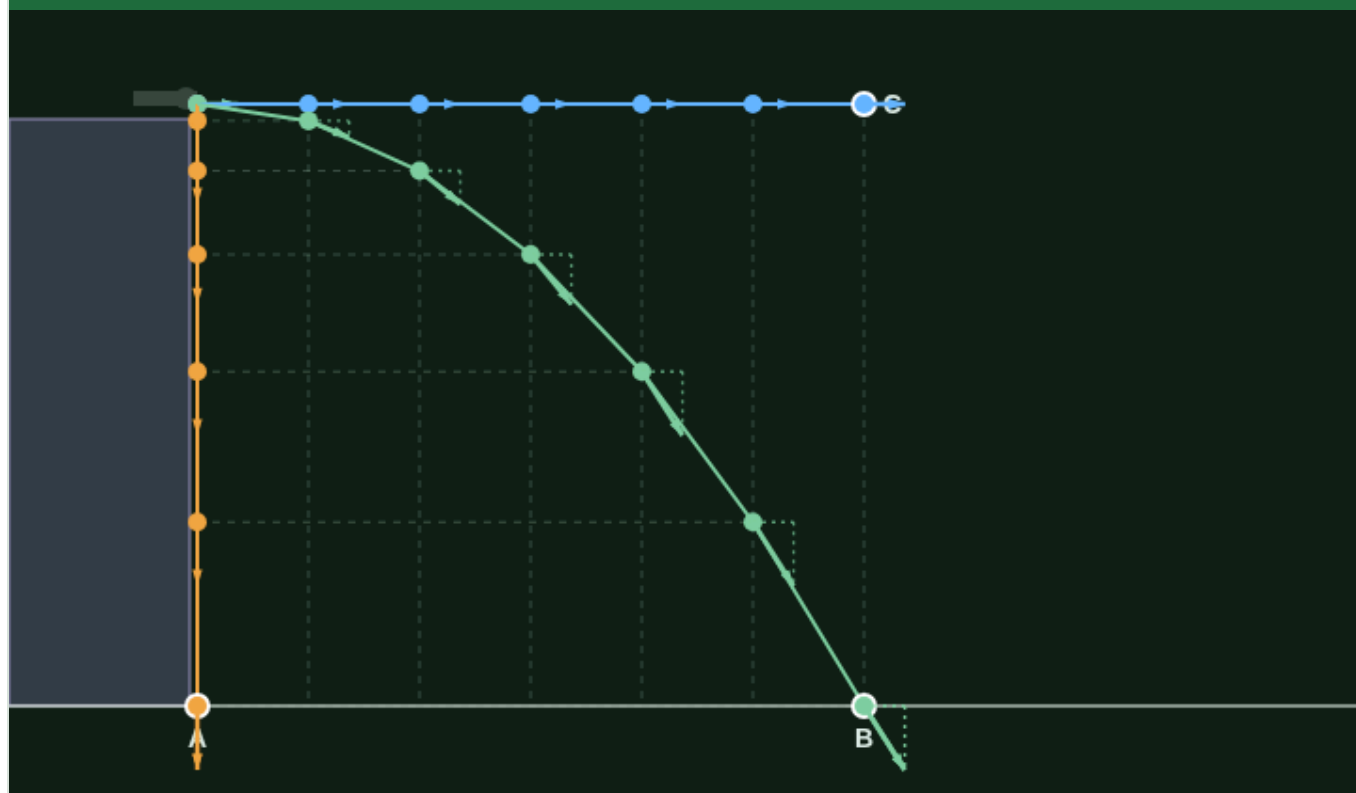
What you should notice: however hard the launcher fires the horizontal ball, you hear a single "thud" when the balls land — not two. Both balls reach the floor at the same instant, every time, no matter the launch speed.

Play with it: trajectories A, B, and C

Trajectory **A** is the dropped ball, **B** is the horizontally launched ball, and **C** is a hypothetical trajectory for the launched ball if it were fired in a gravity-free region of space. Press play and

watch the shared clock — notice which two balls always reach the bottom together, and which one never does. Try the launch-speed slider, then press play again.

► INTERACTIVE · FREEFALL VS. HORIZONTAL LAUNCH



Launch speed v_{0x} 12.0 m/s



► Play

Show velocity vectors

↺ Reset

$t = 1.50$ s

$v_x \approx 12.0$ m/s

$v_y \approx 14.7$ m/s

Landed — at $t = 1.50$ s, both A and B reach the ground together. C is still flying, off to the right.



Explain: reading the vectors

Every arrow you toggled on above is a velocity vector, and together they tell the whole story. Path A's arrow only ever points straight down, and it keeps growing — that's gravity acting alone, accelerating the ball downward at 9.8 m/s^2 with nothing pushing it sideways. Path C's arrow only ever points sideways, and it never changes length — that's what horizontal motion

looks like with no gravity at all: constant velocity, forever. Path B's arrow is just those two combined, added component by component — and the dashed guides show it directly: B's horizontal piece exactly matches C at every instant, and B's vertical piece exactly matches A at every instant. Neither component knows the other exists.

Read the marks, not only the arrows. The marks along path C are laid down at *equal time intervals*, and the gap between them never changes — equal spacing in equal times is exactly what a constant v_x looks like, and B's horizontal spacing matches it mark for mark, all the way down. Along path A the gaps grow from one mark to the next, and the arrow beside them always points straight down and always gets longer — a velocity changing at a steady rate, with nothing sideways about it. B simply inherits both: C's spacing across, A's spacing down.

That's the **independence of motion**: a projectile's horizontal velocity and vertical velocity change (or don't change) completely independently of each other. For the horizontally launched ball, we can write that as a single vector equation:

$$\vec{v}(t) = v_{0x} \hat{i} + (-9.8 t) \hat{j}$$

Where this holds: a **horizontal launch only** ($v_{0y} = 0$), with **up taken as positive**, so the acceleration is $a = -g = -9.8 \text{ m/s}^2$ and $g = 9.8 \text{ m/s}^2$ stays a positive magnitude; air resistance ignored; t measured from the instant of release. **Where the restriction lifts:** launch at an angle and the vertical term starts with a head start of its own — $\vec{v}(t) = v_0 \cos \theta \hat{i} + (v_0 \sin \theta - 9.8t) \hat{j}$ — which is **Note 4**, not this note.

The horizontal component, v_{0x} , never appears with a t — it's constant. The vertical component, $-9.8t$, grows steadily with time — the same acceleration acting on the dropped ball. Set v_{0x} to zero and this is exactly Path A's velocity. That's the reasoning behind the claim: since vertical motion (the only thing that determines *when* a ball reaches the ground) is identical for A and B, they must land at the same instant — no matter how fast the ball is launched sideways.

So how long is it in the air? Because the launched ball starts with no vertical velocity at all, its vertical motion *is* a plain drop. Measuring h as the distance fallen — a positive magnitude, so g enters as 9.8 without its sign — the Module 1 result carries straight over:

$$h = \frac{1}{2} g t^2 \rightarrow t = \sqrt{2h/g}$$

Where this holds: released from rest in the vertical direction ($v_{0y} = 0$), which covers both a plain drop and a *horizontal* launch; h is the distance fallen, taken as a positive magnitude, with $g = 9.8 \text{ m/s}^2$; no air resistance. **Where the restriction lifts:** a launch at an angle starts with $v_{0y} \neq 0$, and the flight time is then found from the full quadratic — **Note 4**.

There is no v_{0x} anywhere in it. Fire one ball off a 20 m tower at 2 m/s and another at 200 m/s and both are in the air for $\sqrt{2 \cdot 20/9.8} \approx 2.0 \text{ s}$ — the fast one simply lands a great deal farther out. *How far* out comes from the same clock, at constant horizontal speed: $x = v_{0x}t$. One clock, two motions, and only one of them decides when the ball lands.

This is Lab 3 — the equation you just built is not only the answer to the cannonball question; it is the whole method of the next lab. In **Lab 3: Projectile Motion** you will fire a ball horizontally off a launcher, measure where it lands, and run $t = \sqrt{2h/g}$ in both directions. **Backwards:** you know h and you measured d , so $v_0 = d / \sqrt{2h/g}$ — a launch speed you never timed and have no stopwatch fast enough to catch. **Forwards:** with v_0 in hand, the same equation predicts d at any new height you choose, and you find out whether the prediction survives contact with a steel ball and a sheet of carbon paper.

Take this question to the lab, and do not settle it here. Keep the launch speed the same and *double the launch height*. Does the ball land twice as far out? You now have everything you need to reason it through — the fall time comes from the height alone, and the horizontal distance is that time at constant speed. Write your prediction down before you measure anything.

Looking ahead — everything on this page is the *horizontal* launch, the case where the ball starts with no vertical velocity at all. That is deliberate: it is the one case where the fall is identical to a plain drop, which is what makes the claim testable. **Note 4** tilts the launcher, and the same two independent motions then run on one shared clock — the reasoning you build here is the reasoning you will use there.

Watch: working through the note

WALKTHROUGH VIDEO • ~10 MIN

Video embed goes here — print the guided notes below before you press play.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. The idea that a projectile's horizontal and vertical motions don't affect each other is called...

- ☐ Newton's third law.
- ☐ The independence of motion.
- ☒ Conservation of momentum.
- ☐ Terminal velocity.

Q2. In a CER (Claim–Evidence–Reasoning) response, the "Reasoning" part should...

- ☐ Restate the claim in different words.
- ☐ List observations without interpreting them.

☐ Connect the evidence to the claim using a scientific principle.

☐ Predict a new, unrelated experiment.

Q3. A ball is dropped from a 20 m tower at the same instant a second ball is fired horizontally from the same height at 15 m/s. Ignoring air resistance, how long does it take the launched ball to hit the ground? (Hint: which velocity component actually controls this?)

s

Q4. In $\vec{V}(t) = v_{0x} \hat{i} + (-9.8t) \hat{j}$ for the launched ball, the term $-9.8t$ represents...

☐ The constant horizontal velocity.

☐ The increasing downward vertical velocity due to gravity.

☐ The total speed of the ball.

☐ The horizontal position at time t .

Check my answers

Clear

Your notes

BEFORE CLASS

Guided notes — blank (PDF)

Print or open on a tablet and work through the CER exercise as you follow along.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.

